Advanced Cluster Management for Kubernetes: Red Hat OpenShift with NetApp

NetApp Solutions

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June 14, 2021

This PDF was generated from https://docs.netapp.com/us-en/netapp-solutions/containers/rh-os-n_use_case_advanced_cluster_management_overview.html on July 11, 2021. Always check docs.netapp.com for the latest.

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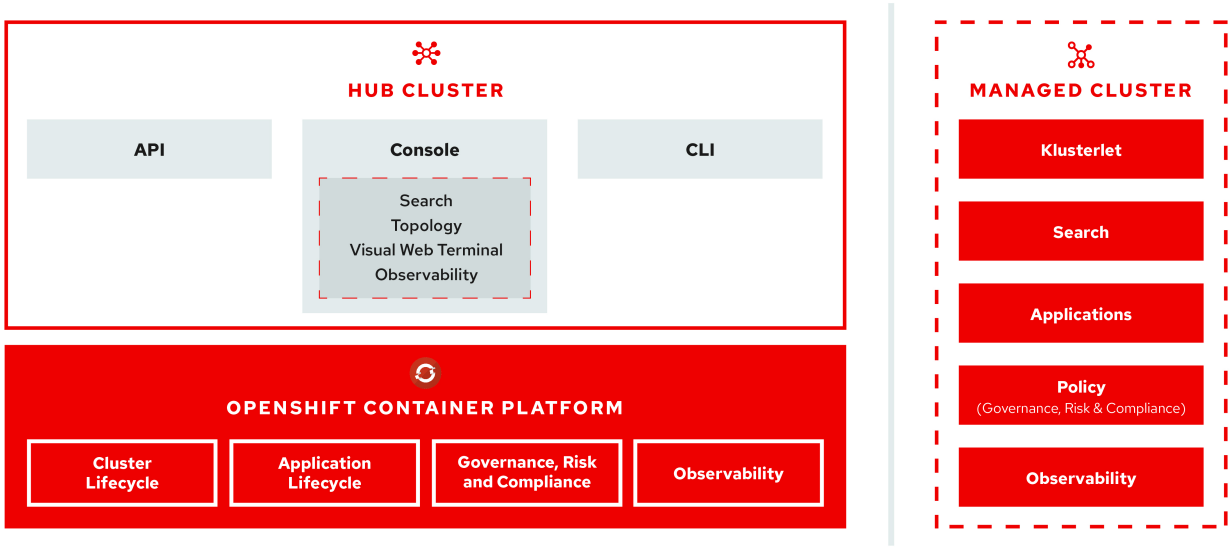
As a containerized application transitions from development to production, many organizations require multiple Red Hat OpenShift clusters to support the testing and deployment of that application. In conjunction with this, organizations usually host multiple applications or workloads on OpenShift clusters. Therefore, each organization ends up managing a set of clusters, and OpenShift administrators must thus face the added challenge of managing and maintaining multiple clusters across a range of environments that span multiple on-premises data centers and public clouds. To address these challenges, Red Hat introduced Advanced Cluster Management for Kubernetes.

Red Hat Advanced Cluster Management for Kubernetes allows the users to:

1. Create, import, and manage multiple clusters across data centers and public clouds.
2. Deploy and manage applications or workloads on multiple clusters from a single console.
3. Monitor and analyse health and status of different cluster resources.
4. Monitor and enforce security compliance across multiple clusters.

Red Hat Advanced Cluster Management for Kubernetes is installed as an add-on to a Red Hat OpenShift cluster, and it uses this cluster as a central controller for all its operations. This cluster is known as hub cluster, and it exposes a management plane for the users to connect to Advanced Cluster Management. All the other OpenShift clusters that are either imported or created via the Advanced Cluster Management console are managed by the hub cluster and are called managed clusters. It installs an agent called Klusterlet on the managed clusters to connect them to the hub cluster and serve the requests for different activities related to cluster lifecycle management, application lifecycle management, observability, and security compliance.



For more information, see the documentation here.

Next: Deployment Prerequisites.

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